16·05·2025 (M)

SET-F

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Semester: January 2025– April 2025		
Maximum Marks: 50 Examination: End-Semester Examination Duration: 2 Hrs. 30 min		
Programme code: H05 Programme: Honors Software and Mobile Application Development	Class: S.Y.	Semester: IV (SVU 2023)
Institute/School/Department: K. J. Somaiya School of Engineering	Name of the department: COMPUTER	
Course Code: 216H05C401	Name of the Course: Fundamental of Swift	
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question Statement	Max. Marks
Q.1	Attempt any **four**	20
a	Write a Swift struct that conforms to the Comparable protocol and explain its significance.	05
b	Explain Protocol Associated Types in Swift.	05
c	How would you write JSON data to a file and read it back in Swift?	05
d	How does RawRepresentable protocol help in working with enums? Write an enum conforming to it.	05
e	Write a Swift class that inherits from a generic class. Explain the use case.	05
f	Describe how operators can be treated as functions in Swift. Provide an example explaining the concept.	05
Q.2	Attempt any **One**	10
a	• Create a User class with optional email and phoneNumber properties. • Write a method displayContactInfo() that uses guard let and if let to unwrap values. • If values are missing, provide default messages using the?? Operator. • Also include a required name property and use guard to ensure it's not empty. • Demonstrate with different user objects: full info, partial info, and empty name.	
b	A digital library recommends books based on the number of times they've been read. Requirements: • Create a class Book with properties title, author, and readCount (with didSet to print a message when updated). • Store multiple Book objects in an array. • Write a function that accepts a closure and returns all books read more than a certain number of times. • Demonstrate using that closure to get recommended books.	

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Q.3	Attempt the following	20
	Design a gallery app: • Main screen has 3 buttons with image titles. • Tapping a button navigates to a second screen showing the selected image and title using a `UIImageView` and `UILabel`. **Requirements:** • Use `UIButton` and `UIImageView`. • Use **Show Segue** for navigation. • Use **prepare(for:sender:)** to pass selected image name. • Apply Auto Layout to scale image appropriately on second screen.	